PVP 20

Code: 20CS3601

III B.Tech - II Semester – Regular / Supplementary Examinations
APRIL 2024

COMPILER DESIGN
(COMPUTER SCIENCE & ENGINEERING)

Duration: 3 hours Max. Marks: 70

Note: 1. This paper contains questions from 5 units of Syllabus. Each unit carries 14 marks and have an internal choice of Questions.
2. All parts of Question must be answered in one place.

BL – Blooms Level CO – Course Outcome

			BL	CO	Max. Marks
		UNIT-I			
1	a)	Differentiate tokens, patterns, lexeme.	L2	CO1	7 M
	b)	Write a short note on: i. Assembler ii. Linkers and loaders	L2	CO1	7 M
		OR			
2	a)	What are the various phases of a compiler? Explain each phase in detail by using the input “x=(y+z)*2”.	L2	CO1	10 M
	b)	What are the various parts in ***LEX*** program?	L2	CO1	4 M
		UNIT-II			
3	a)	Write the production rules to eliminate the left recursion and left factoring problems.	L2	CO1	4 M

	b)	Consider the grammar. S->AB|ABad A->d E ->b D->b| ε B->c Construct the predictive parsing table. Show that the given grammar is *LL(1) or not.*	L3	CO2	10 M
OR					
4	a)	Write Rules to construct *FIRST* function and *FOLLOW* function.	L2	CO1	4 M
	b)	Show the following Grammar S-> AaAb|BbBa A->ε B->ε Is ***LL(1)*** and parse the input string ***"ba".***	L3	CO2	10 M
UNIT-III					
5	a)	What is a shift-reduce parser? Explain in detail the conflicts that may occur during shift-reduce parsing.	L2	CO1	4 M
	b)	Write the rules to construct the SLR parsing table. E-> E+T|T T-> TF|F F-> F*|a|b Construct the SLR parsing table and also parse the input "a*b+a".	L2	CO3	10 M

OR					
6	a)	Write the rules to construct the SLR parsing table.	L2	CO1	7 M
	b)	List all *LR(0)* items for the following grammar: S→AS | b A→SA | a	L3	CO3	7 M
UNIT-IV					
7	a)	What is LALR parser? Construct the set of LR(1) items for this grammar: S-> CC C-> aC C->d	L3	CO3	7 M
	b)	What is intermediate code and write the two benefits of intermediate code generation.	L2	CO4	7 M
OR					
8	a)	Draw the syntax tree and ***DAG*** (Direct Acyclic Graph) for the following expression: *(a*b)+(c-d)*(a*b)+b*	L3	CO4	7 M
	b)	i. What are the limitations of static allocation? ii. List Dynamic Storage allocation techniques.	L4	CO5	7 M

UNIT-V					
9	a)	What are basic blocks? What is the use of algebraic identities in optimization of basic blocks?	L2	CO4	7 M
	b)	What do you mean by machine dependent and machine independent optimization?	L3	CO1	7 M
OR					
10	a)	Write a short note with example to optimize the code: i. Dead code elimination. ii. Code motion.	L3	CO1	7 M
	b)	Discuss the various peephole optimization techniques in detail.	L2	CO4	7 M

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UNIT-I					
1	a)	Explain the role of lexical analysis in the compilation process. How does it interact with other phases of a compiler?	L2	CO1	7 M
	b)	Explain the working of the Lexical Analyzer Generator (LEX). How does it simplify the development of a lexical analyzer? Provide a sample LEX program.	L2	CO1	7 M
OR					
2	a)	Discuss the various phases of a compiler and explain the role of each phase in the compilation process.	L2	C01	7 M
	b)	Define tokens, lexemes, and patterns in the context of lexical analysis. How are	L2	C01	7 M

		tokens specified and recognized by a lexical analyzer?			
UNIT-II					
3	a)	Discuss the steps involved in writing a grammar for a programming language construct with an example.	L2	C02	4 M
	b)	Explain the concept of LL(1) grammar. What are the conditions for a grammar to be LL(1), verify with the following example. $E \rightarrow TE^{I}$ $E^{I} \rightarrow +TE^{I} \mid \varepsilon$ $T \rightarrow FT^{I}$ $T^{I} \rightarrow *FT^{I} \mid \varepsilon$ $F \rightarrow (E) \mid id$	L3	C02	10 M
OR					
4	a)	Compare the process of non-recursive predictive parsing versus recursive descent parsing.	L4	C05	7 M
	b)	What are the common types of errors encountered in predictive parsing and how are they handled during error recovery?	L3	C02	7 M
UNIT-III					
5	a)	What is handle pruning in bottom-up parsing? Describe the process and its significance in constructing the parse tree.	L2	C03	5 M
	b)	Explain the shift-reduce parsing technique. Illustrate its operations with	L3	C03	9 M

		the help of following example. S→(L) | a L→L,S | S			
OR					
6	a)	Describe the construction of an LR(0) automaton. Illustrate the process with the following example. S→AS | b A→SA | a	L3	C03	7 M
	b)	Discuss the LR parsing algorithm, explaining each step in the parsing process with the following example. S→AS | b A→SA | a	L3	C03	7 M
UNIT-IV					
7	a)	Explain the construction of a Canonical LR(1) parsing table. Demonstrate the process with the following grammar. S→CC C→aC | d	L3	C03	10 M
	b)	What is an LALR parser, and how does it differ from Canonical LR(1) parsers? Why is LALR parsing preferred in certain scenarios?	L4	C05	4 M
OR					
8	a)	Explain the organization of storage in a runtime environment. Discuss the key components involved in storage organization?	L3	C04	7 M

	b)	Generate three address code for the following expression. (a*b) + (c+d) – (a+b+c)	L3	C04	7 M
UNIT-V					
9	a)	Discuss the advantages of using a Directed Acyclic Graph (DAG) to represent a basic block. How does DAG representation facilitate optimization?	L3	C04	7 M
	b)	What are the principle sources of optimization in machine-independent code optimization? Provide specific examples of each source.	L3	C04	7 M
OR					
10	a)	Describe the concept of "Semantic Preserving Transformations." Why are these transformations crucial in code optimization?	L2	C01	7 M
	b)	Explain the difference between "Local Common Sub-expressions" and "Global Common Sub-expressions". How are these redundancies eliminated during optimization?	L3	C04	7 M